

Short communication

MLCT interaction in $[\text{Rh}(\text{II})_2(\text{CO}_3)_4]^{4-} (\times \text{L})$

Arnd Vogler

Institute of Inorganic Chemistry, University Regensburg, Germany



ARTICLE INFO

Article history:

Received 16 July 2015

Received in revised form 2 October 2015

Accepted 9 October 2015

Available online xxx

Keywords:

Photochemistry

Rhodium(II)

Carbonate

Photoreduction of CO_2

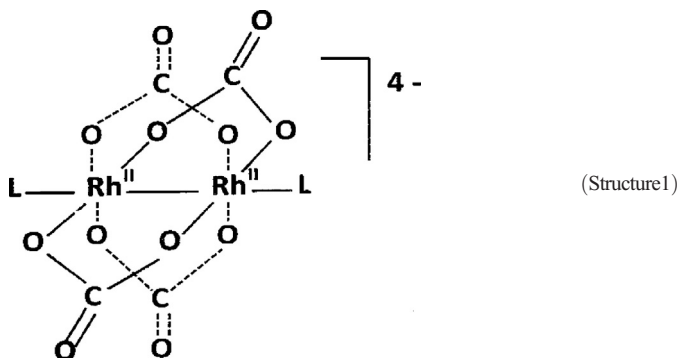
MLCT excitation

ABSTRACT

Complexes of the type $[\text{Rh}^{II}_2(\text{CO}_3)_4(\text{H}_2\text{O})\text{L}]^{n-}$ with $\text{L} = \text{N-methylpyrazinium}^+$ and 1-heptyl-4-(4-pyridinyl)pyridinium $^+$ cations display intense long-wavelength ($\text{Rh}(\text{II})$ to L) MLCT absorptions. With $\text{L} = \text{H}_2\text{O}$, MLCT absorptions are not identified, but the photoreactivity of the complex in aqueous solution supports the assumption that ($\text{Rh}(\text{II})$ to CO_3^{2-}) MLCT excited states are accessible. Upon irradiation with white light, $\text{Rh}(\text{II})$ is photooxidized while carbonate is reduced to CO . The efficiency of this photolysis is very low. However, the occurrence of this photoredox reaction is, nevertheless, of general interest with regard to the photochemical reduction of CO_2 .

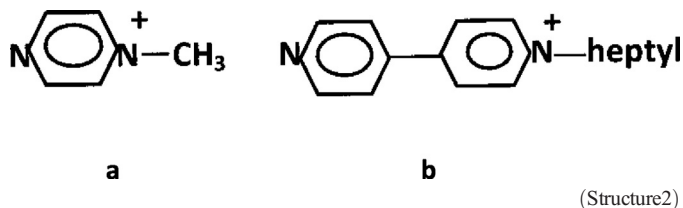
© 2015 Published by Elsevier B.V.

The photoreduction of carbon dioxide plays an essential role in natural photosynthesis. Various models have been developed in order to imitate this process. An artificial photosystem of this type may take advantage of suitable metal complexes as primary electron donors [1,2]. As electron acceptor, CO_2 could be replaced by carbonate [3], which serves as a rather stable and versatile ligand. The simplest reduction of CO_3^{2-} involves the acceptance of 2 electrons yielding CO as energy-rich product. Both electrons may originate from a reducing metal which can be oxidized by 2 units. As an alternative, both electrons may be supplied by two metal centers of binuclear complexes. Recently, we have investigated copper(I) complexes as photoreductants for carbonate [3,4]. In the present study, we examined the anion $[\text{Rh}^{II}_2(\text{CO}_3)_4]^{4-}$ [5] (structure 1) [6] as a suitable candidate.



This binuclear complex contains $\text{Rh}(\text{II})$ as electron donor and carbonate as electron acceptor. However, the acceptor orbitals of the carbonate ligands are certainly located at very high energies as discussed before [7]. Accordingly, MLCT excitation terminating at carbonate,

may require very short-wavelength irradiation which might not be available by conventional light sources. Although $\text{Rh}(\text{II})$ is reducing, the metal–metal bonding of this binuclear complex could modify its electron-donating ability since the d-orbital derived HOMO may be stabilized by meta–metal bonding. This can be probed by attaching appropriate CT-acceptor ligands L in the axial position. The best ligands of this type are cations such as $\text{N-methylpyrazinium}^+$ and 1-heptyl-4-(4-pyridinyl)pyridinium $^+$ (structure 2a and b). The salt $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4]$ was prepared by a literature procedure [5]. Its absorption spectrum (Fig. 1) agrees very well with that previously reported [5].



(Structure 2)

Upon addition of $\text{N-methylpyrazinium}$ perchlorate and $\text{N-heptyl-4,4'-bipyridyl}$ bromide to an aqueous solution of $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4]$, the light blue solution changed to dark blue and dark wine-red, respectively. From the dark blue solution the salt $\text{Na}_3[\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})(\text{N-methylpyrazinium})]$ was precipitated by addition of ethanol, washed with ethanol, ether and dried under vacuum. The salt was obtained as dark blue substance (yield: approximately 80%). Anal. Calc. (F.W. 628) C 17.2, H 1.4, N 4.5. Found: C 16.9, H 2.9, N 4.9. ESI-MS for $[\text{Rh}_2(\text{CO}_3)_4(\text{N-CH}_3\text{-pyrazinium})]^{3-}$ ($\text{C}_9\text{H}_9\text{N}_2\text{O}_{12}\text{Rh}_2$): $m/z = 543$ (complex + 2H^+), 542 (complex + H^+), in agreement with the calculation.

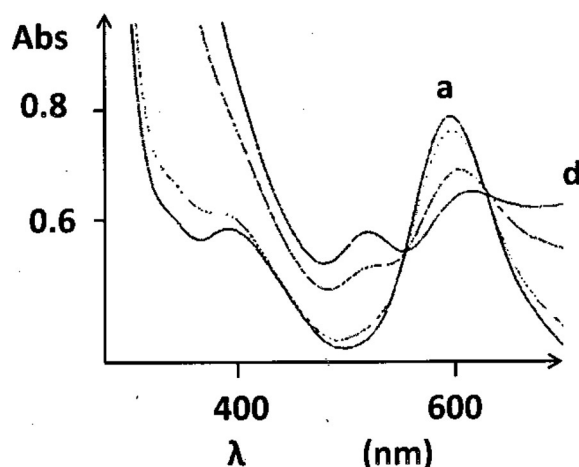


Fig. 1. Spectral changes during the irradiation of 3.7×10^{-3} M $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4]$ in water at r.t. in a 1-cm quartz cell with white light. Irradiation time: (a) 0 min, 10 min, 130 min, (d) 310 min.

Irradiations were performed with a high-pressure mercury lamp Osram 100 W/2.

The absorption spectrum of $\text{Na}_3[\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})(\text{N-methylpyrazinium})]$ in water (Fig. 2) shows an intense long-wavelength band at $\lambda_{\text{max}} = 625$ nm, $\epsilon = 2370 \text{ M}^{-1} \text{ cm}^{-1}$ which displays a distinct shoulder above 500 nm. An additional band appears at $\lambda_{\text{max}} = 359$ nm. The salt $\text{Na}_3[\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})(1\text{-heptyl-4-(4-pyridinyl)pyridinium})]$ was not isolated. Its wine-red solution in water exhibits its longest-wavelength absorption (Fig. 2) at $\lambda_{\text{max}} = 465$ nm ($\epsilon = 2970$). The 500 nm shoulder appears again, but this time at the long-wavelength side of the 465 nm maximum.

In an aqueous solution the anion $[\text{Rh}_2(\text{CO}_3)_4]^{4-}$ exists as $[\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})_2]$ [5] ($\text{L} = \text{H}_2\text{O}$ in Structure 1). This blue solution is only slightly light-sensitive. Upon irradiation with white light for several hours under argon, a photolysis takes place which is accompanied by spectral changes (Fig. 1) including two isosbestic points at 550 and 626 nm and a decrease of the optical density at

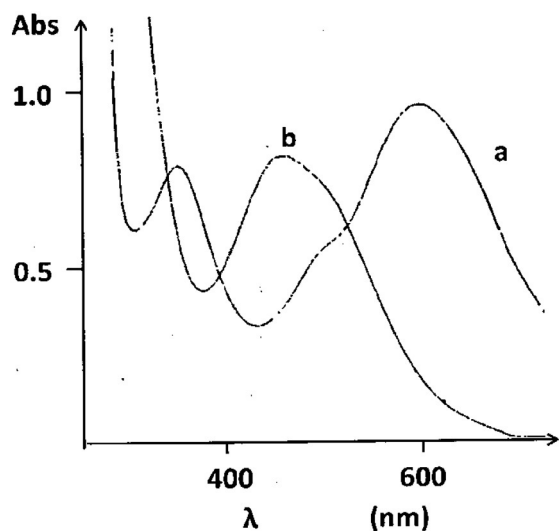


Fig. 2. Absorption spectrum of $[\text{Rh}_2(\text{CO}_3)_4(\text{H}_2\text{O})\text{L}]^{3-}$ in a 1-cm cell at r.t.; 4.2×10^{-4} M for $\text{L} = \text{N-methylpyrazinium}^+$ (a) and 2.6×10^{-4} M for $\text{L} = 1\text{-heptyl-4-(4-pyridinyl)pyridinium}^+$ (b).

the absorption maximum at 600 nm. It was assumed that the binuclear rhodium (II) complex underwent a photooxidation with the loss the 600 nm absorption which is associated with Rh(II)-Rh(II) bond. A thermal oxidation of the related binuclear acetate complex $[\text{Rh}^{\text{II}}_2(\text{acetate})_4]$ to Rh(III) takes place, when a basic peroxide is added as an oxidant [8]. Indeed, upon addition of H_2O_2 to an aqueous solution of $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4]$, spectral changes are observed (Fig. 3) which closely resemble those during the photolysis of the complex anion (Fig. 1). Upon further addition of H_2O_2 the spectral pattern becomes more complicated indicating additional side and secondary reactions. Nevertheless, the close resemblance of Figs. 1 and 3 and the analogous reaction of $[\text{Rh}_2(\text{acetate})_4]$ with peroxide [8] clearly indicate the photooxidation of $[\text{Rh}^{\text{II}}_2(\text{CO}_3)_4]^{4-}$ to Rh(III) .

What is the concomitant product of the photoreduction? Only hydrogen or carbon monoxide are reasonable candidates. Finally, we succeeded to detect CO with Dräger tubes which are based on the reaction: $5 \text{ CO} + \text{I}_2\text{O}_5 \rightarrow \text{I}_2 + 5 \text{ CO}_2$. Owing to the very slow photolysis of aqueous $[\text{Rh}_2(\text{CO}_3)_4]^{4-}$ and the occurrence of side and/or secondary photoreactions at later stages of the photolysis as indicated by the deviation from the isosbestic points (see Fig. 1), any further products could not be identified. The pH does not change upon irradiation.

The electronic structure and the electronic absorption spectra of binuclear Rh(II) complexes, in particular $\text{Rh(II)}_2(\text{acetate})_4$, have been extensively discussed [9–12]. Under consideration of the presence of the metal–metal bonding, low-energy electronic transitions are associated with the metal–metal bonding MOs and LMCT excitation. The characteristic long-wavelength absorption near 600 nm is associated with the Rh(II)-Rh(II) bond. Such absorptions are not very intense with extinction coefficients of approximately 100 [5,8,12]. In the case of $[\text{Rh}^{\text{II}}_2(\text{CO}_3)_4\text{L}]^{n-}$, the significance of MLCT transitions was not obvious, but is now rather clear. The complex anions $[\text{Rh}^{\text{II}}_2(\text{CO}_3)_4(\text{H}_2\text{O})\text{L}]^{3-}$ with $\text{L} = \text{N-methylpyrazinium}^+$ and 1-heptyl-4-(4-pyridinyl)pyridinium $^+$ display intense (Rh^{II} to L) MLCT absorptions at $\lambda_{\text{max}} = 625$ and 475 nm, respectively. In the case of $[\text{Fe}^{\text{II}}(\text{CN})_5\text{L}]^{2-}$, the corresponding MLCT absorptions have been reported to appear at 661 and 533 nm [13]. When $\text{L} = \text{H}_2\text{O}$, low-energy MLCT bands are apparently absent. However, (metal to π^* carbonate) MLCT transitions must occur at rather high energies [7], but in the UV region there is certainly an accumulation of absorptions of quite different origin. The photolysis of $[\text{Rh}^{\text{II}}_2(\text{CO}_3)_4]^{4-}$ in aqueous solution may be taken as support for the accessibility of a photoactive MLCT state. The very low efficiency is certainly

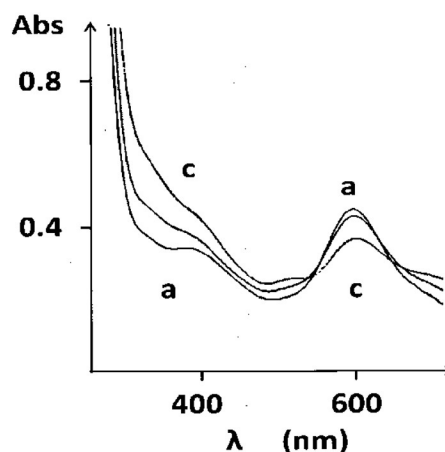
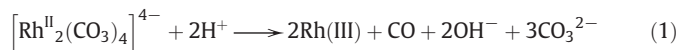


Fig. 3

Fig. 3. Spectral changes during the reaction of 1×10^{-3} M $\text{Na}_4[\text{Rh}_2(\text{CO}_3)_4]$ in water (a) with increasing amounts of H_2O_2 (b and c), at r.t. in a 1-cm quartz cell.

associated with the presence of various competing deactivation steps. In view of our observations, we suggest that the photolysis proceeds according to Eq. 1.



Free carbonate and hydroxide were not detected, but may be attached to rhodium(III). The complex $[\text{Rh}_2(\text{CO}_3)_4(\text{N-methyl pyrazinium})]^{3-}$ was not affected by long-wavelength irradiation (>500 nm) into the 625 nm band. Generally, this type of MLCT excitation is rapidly reversed by a facile back electron transfer.

In conclusion, the $\text{Rh}^{\text{II}}\text{--Rh}^{\text{II}}$ bond has been shown to serve as a CT-donor. In the case of $[\text{Rh}^{\text{II}}_2(\text{CO}_3)_4]^{4-}$, a (Rh^{II} to π^* carbonate) MLCT state seems to be present at high energies. MLCT excitation leads to a photolysis with a rather low efficiency. The results are consistent with an oxidation of $\text{Rh}(\text{II})$ and the reduction of carbonate to CO. Our study

may be a valuable contribution to the general problem of the photochemical reduction of carbon dioxide.

References

- [1] A.J. Morris, G.J. Meyer, E. Fujita, *Acc. Chem. Res.* 42 (2009) 1983–1994.
- [2] H. Takeda, O. Ishitani, *Coord. Chem. Rev.* 254 (2010) 346–354.
- [3] H. Kunkely, A. Vogler, *Inorg. Chem. Commun.* 13 (2010) 137–138.
- [4] A. Vogler, H. Kunkely, *Inorg. Chem. Commun.* 14 (2011) 96–98.
- [5] C.R. Wilson, H. Taube, *Inorg. Chem.* 14 (1975) 405–409.
- [6] F.A. Cotton, T.R. Felthouse, *Inorg. Chem.* 19 (1980) 320–323.
- [7] M. Leirer, G. Knör, A. Vogler, *Inorg. Chem. Commun.* 1 (1998) 49–50.
- [8] C.R. Wilson, H. Taube, *Inorg. Chem.* 14 (1975) 2276–2279.
- [9] L. Dubicki, R.L. Martin, *Inorg. Chem.* 9 (1970) 673–675.
- [10] J.G. Norman, H.J. Kolari, *J. Am. Chem. Soc.* 100 (1978) 791–799.
- [11] B.E. Bursten, F.A. Cotton, *Inorg. Chem.* 20 (1981) 3042–3048.
- [12] V.M. Miskowski, W.P. Schaefer, B. Sadeghi, B.D. Santarsiero, H.B. Gray, *Inorg. Chem.* 23 (1984) 1154–1162.
- [13] B.J. Coe, S.P. Foxon, E.C. Harper, J. Raftery, R. Shaw, C.A. Swanson, I. Asselberghs, K. Clays, B.S. Brunshwig, A. Fitch, *Inorg. Chem.* 48 (2009) 1370–1379.